

HW03-AI — GUI & Usability Testing on EMS

1. Cover and Metadata

Field	Value
Student	Đinh Hoàng Nam
Student ID	23127430
Class / Cohort	23KTPM1
Course	CS423 / CSC13003 — Software Testing
Assignment	HW03-AI — GUI & Usability Testing on EMS
AI agent	OpenAI Codex
Scenario	Scenario C — Admin manages users
Owned screens	Exactly C1, C2, and C3
C1	Users List — search, role/active filters, and user columns
C2	Assign Role / Edit User
C3	Block/Unblock and Reset Password dialogs — confirmation and audit feedback
Current SUT	https://prod-dev.ems-fitus.cloud/
Report status	Content synchronized; remaining finalization actions are Google Form submission, one skill-demo video, final Git log, PDF export, and ZIP packaging
Test dates	Task 1B: 2026-08-02, 13:08:06–14:54:02 (UTC+07:00); Task 2 official sessions: 2026-08-02 to 2026-08-03; Task 3: 2026-08-04
Test environment	Task 1B: authenticated Codex in-app browser, 510 × 698 observed viewport; Task 3: physical Windows, iPadOS, and Android devices
Student evidence overlay	dhnam23@clc.fitus.edu.vn with student/device identification

Task 1B reports a completed authorised live EMS execution with genuine SUT screenshots. Task 2 reports one pilot, five official sessions, corrected aggregate analysis, and SUS-derived conclusions. Task 3 contains 15 executed physical-device cells and one confirmed Android responsive-layout failure. The only remaining actions are the Google Form, one skill-demo video, the final Git log, PDF export, and ZIP packaging/submission.

Source discrepancy

The official brief and the student-supplied scope both specify <https://prod-dev.ems-fitus.cloud/>; there is no conflicting URL value. The brief also calls the service an ngrok tunnel. That infrastructure statement is not demonstrated by the supplied hostname and is retained only as historical course text. The current SUT URL above governs this working scope.

2. Executive Test Summary

Scenario C is an end-to-end user-administration flow across C1–C3. The group approved all 56 checklist items, Task 1B live execution is complete, the pilot and five official Task 2 sessions have been analysed, and all 15 Task 3 compatibility cells have been executed.

Measure	Current evidence-backed value
Shared checklist	56 preserved AI Draft items: 14 each in IA-01...IA-04; approved unchanged
Human checklist review/additions	56/56 group-approved; 0 AI-Assisted Review additions; 0 Human Added
Checklist screen-item cells	168 executed; 150 Passed; 18 Failed; 0 BLOCKED
Passed / Failed	150 / 18
Verified bugs	10 total: 9 from Task 1B and 1 from Task 3
Evidence-required bug candidates	0; Reset Password absence is confirmed as C3-RESET-001

Measure	Current evidence-backed value
Pilot / official participants	1 / 5 documented; pilot achieved a 100% scenario-comprehension pass
Official-session metrics	100% available-task success; mean time 60.0 s; mean errors/hesitations/assistance 0.0; Reset Password system-blocked for 5/5
Mean SUS	68.5 / 100
Task 2 usability conclusion	5/5 participants were system-blocked by existing bug C3-RESET-001 ; no separate usability-finding ID is counted
Compatibility cells	15 planned and executed; 14 Passed, 1 Failed (C1-RESPONSIVE-001)
Skill demo videos	1 required for ems-checklist-executor ; link pending

3. Scope, Screen Selection, and Exclusions

C1 provides the user inventory and identification/filtering context; C2 changes the selected user's role; C3 controls account access and password recovery with confirmation/audit feedback. Together they form one coherent administrator goal: locate a user, change authority, manage access, and verify system state.

C4 Export to Excel is explicitly excluded. Task 2 starts at **Admin** → **Users List**. Switching from the ordinary user interface to the admin interface is pre-task setup and is outside the measured C1–C3 task unless separately timed and reported.

Compatibility uses only Windows, iPadOS, and Android. macOS is outside the available environment and is not tested.

4. Task 1A — Shared GUI Checklist Method

4.1 Draft, sources, and provenance

The supplied workbook had a broken external reference to [\[1\]Master Checklist](#). Its cached external-link data retained 56 checklist rows, which were recovered without changing their wording. The student has since confirmed that the group reviewed and approved all 56 rows unchanged. The distribution is:

Interface aspect	Items
IA-01 General UI	14
IA-02 Forms	14
IA-03 Navigation	14
IA-04 Feedback / State	14
Total	56

The authoritative row-level checklist is stored in [shared_gui_checklist.md](#). The supplied workbook content was migrated to Markdown and the spreadsheet files were then removed at the student's request. References cover the course slides, Nielsen, Norman, Shneiderman, W3C/WCAG, SUS, ISTQB, BrowserStack, and EMS materials; see [reference_sources.md](#).

4.2 Origin labels

- **AI Draft**: one of the 56 preserved original rows.
- **AI-Assisted Review**: any later item or rewrite proposed by an AI tool, even if accepted or edited by a student.
- **Human Added**: only a genuinely student-originated addition, with an explanation of why the prior AI work missed it.

Current counts are 56 AI Draft, 0 AI-Assisted Review additions, and 0 Human Added items. All 56 draft items were group-reviewed and approved unchanged.

4.3 Prompt and review trace

The prior kickoff prompt and reusable review templates are preserved in [checklist_ai_prompts.md](#).

Review subject	Duplicate/rewrite/addition decision	Why AI missed it	Status
56 preserved draft rows	Approved unchanged; no deletion or rewrite reported	Not applicable — no new item was added	Approved 56/56
EMS-specific human addition(s)	None reported	Not applicable — no addition was made	0 additions

Review subject	Duplicate/rewrite/addition decision	Why AI missed it	Status
Accessibility, keyboard, EN/VI, RTL/dark-mode relevance	Covered where represented in the approved draft; no change was requested	Live applicability still depends on Task 1 execution	Approved as part of the 56-row review

5. Task 1B — Checklist Execution on Scenario C

The approved 56-item checklist was executed against all three owned screens on the live SUT, producing 168 final screen-item results with no **BLOCKED** cells.

- SUT: <https://prod-dev.ems-fitus.cloud/>.
- Live run: 2026-08-02, 13:08:06–14:54:02 (UTC+07:00).
- Environment: authenticated Codex in-app browser; observed 510 × 698 viewport.
- Dedicated C2/C3 account: Dinh Nam (hnam23@clc.fitus.edu.vn).
- Restoration: temporary profile, role, and Active-state changes were restored to Dinh Nam, Student, blank Phone/Member Code, and Active. The audit Updated timestamp changed as an expected consequence of the authorised saves.
- Final validator: 56 checklist rows, 168 result cells, 150 Passed, 18 Failed, 0 working statuses; 0 errors and 0 warnings.

Passed records either an applicable behaviour observed without a violation or a criterion whose control/state is not instantiated on that screen. **Failed** records a directly observed violation with a stable bug ID and genuine SUT screenshot in the supporting execution file.

ID	Checklist criterion	C1	C2	C3
IA01-01	Consistent grid/alignment; no overlap or clipping	Passed	Passed	Passed
IA01-02	Responsive on desktop/tablet/phone without hidden actions	Passed	Passed	Passed
IA01-03	Clear, consistent typography hierarchy	Passed	Passed	Passed
IA01-04	Sufficient contrast in all control states	Passed	Passed	Passed
IA01-05	Meaning is not communicated by colour alone	Passed	Passed	Passed
IA01-06	Images/icons are sharp, correctly proportioned, and accessible	Failed	Passed	Passed
IA01-07	Interactive states are visibly distinguishable	Passed	Passed	Passed
IA01-08	EN/VI text is complete and consistent	Passed	Failed	Passed
IA01-09	Locale formats are unambiguous and consistent	Failed	Passed	Passed
IA01-10	Loading/empty/error/denied states are distinct and stable	Passed	Passed	Passed
IA01-11	Terms/icons/roles/statuses match user mental models	Passed	Failed	Passed
IA01-12	Only relevant content/actions compete for attention	Passed	Passed	Passed
IA01-13	Page title, heading, and route identify the screen	Passed	Passed	Failed
IA01-14	Content works with zoom and long translations	Passed	Passed	Passed
IA02-01	Inputs have persistent programmatic labels	Passed	Passed	Passed
IA02-02	Required fields are marked visibly and programmatically	Passed	Passed	Passed
IA02-03	Formats, limits, and examples appear before entry	Passed	Failed	Passed
IA02-04	Form keyboard order and focus indicators are logical	Passed	Passed	Passed
IA02-05	Input type/keyboard/autocomplete/masking fits data	Passed	Passed	Passed
IA02-06	Validation timing does not interrupt entry	Passed	Passed	Passed
IA02-07	Errors are specific, plain, proximal, and not colour-only	Passed	Failed	Passed
IA02-08	Failed submission focuses useful error and identifies all invalid fields	Passed	Failed	Passed
IA02-09	Non-sensitive values survive validation/server errors	Passed	Passed	Passed
IA02-10	Related dates/times enforce ordering and explain conflicts	Passed	Passed	Passed
IA02-11	Upload communicates constraints, preview, progress, and recovery	Passed	Passed	Passed
IA02-12	Rich-text controls are labelled, keyboard-operable, predictable	Passed	Passed	Passed
IA02-13	Selectors/toggles show state and enforce dependencies	Passed	Passed	Passed

ID	Checklist criterion	C1	C2	C3
IA02-14	Submit prevents duplicates, shows progress, explains disabled state	Passed	Passed	Passed
IA03-01	Global navigation is consistent across screens/roles	Passed	Passed	Passed
IA03-02	Current location is visibly indicated	Passed	Passed	Passed
IA03-03	Links/buttons and labels match their navigation/action semantics	Failed	Failed	Failed
IA03-04	Back/Cancel/Close/Return preserve predictable context	Passed	Passed	Passed
IA03-05	Deep links and invalid/unauthorized links recover safely	Passed	Passed	Passed
IA03-06	Tabs/filters show state and preserve meaningful location	Passed	Passed	Passed
IA03-07	Search/filters show labels, state, count, clear, persistence	Failed	Passed	Passed
IA03-08	Essential navigation works by keyboard without trap	Passed	Passed	Passed
IA03-09	Modal/drawer/menu focus moves and restores logically	Passed	Failed	Failed
IA03-10	Overlays support Escape, close/cancel, safe outside click	Passed	Passed	Passed
IA03-11	Drag-and-drop has feedback, recovery, and alternative	Passed	Passed	Passed
IA03-12	Pagination/infinite lists preserve filters/context	Passed	Passed	Passed
IA03-13	External links/downloads are identifiable and non-disruptive	Passed	Passed	Passed
IA03-14	Responsive navigation keeps destinations/actions discoverable	Failed	Passed	Passed
IA04-01	Actions receive timely, proportionate, understandable feedback	Passed	Passed	Passed
IA04-02	Async operations show progress and prevent conflicts	Passed	Passed	Passed
IA04-03	Success feedback states change and next action	Passed	Passed	Passed
IA04-04	Failure feedback explains, preserves state, and supports recovery	Passed	Failed	Passed
IA04-05	Toasts/alerts are readable, dismissible, unobstructive, announced	Passed	Passed	Passed
IA04-06	Status badges are consistent across views	Passed	Passed	Passed
IA04-07	Confirmation identifies object/consequence/action and safe focus	Passed	Passed	Failed
IA04-08	Actions can be cancelled/undone/reversed when reasonable	Passed	Passed	Passed
IA04-09	Progress indicators are accurate/labeled/determinate-aware	Passed	Passed	Passed
IA04-10	Real-time updates preserve context and announce changes	Passed	Passed	Passed
IA04-11	Empty/no-result states explain and offer next action	Failed	Passed	Passed
IA04-12	Auth/session feedback distinguishes states without disclosure	Passed	Passed	Passed
IA04-13	Disabled controls expose unmet prerequisites	Passed	Passed	Passed
IA04-14	Repeats/retries/reconnects do not duplicate or conflict	Passed	Passed	Passed

Screen	Items	Passed	Failed	Blocked	Execution status
C1	56	50	6	0	Final live execution
C2	56	48	8	0	Final live execution
C3	56	52	4	0	Final live execution
Total	168	150	18	0	Final validator passed

Confirmed failure themes are modal focus handling, missing accessible navigation labels, ambiguous locale dates, weak no-result recovery, inconsistent field microcopy and validation, missing Block/Unblock confirmation, and unavailable Reset Password functionality. Detailed reasons and per-cell evidence are in [execution_C1_C2_C3.md](#).

6. Bug Reports from Checklist Execution

Nine genuine Task 1B bugs remain after student review. Two preliminary C1 findings were removed when IA01-07, IA02-04, IA02-13, IA03-06, and IA03-12 were confirmed Passed in practice. Google Form submission has not been claimed.

Bug ID	Screen	Severity	Confirmed actual result	Screenshot evidence	Google Form
C1-LOCALE-001	C1	2 — Minor/medium localisation defect	Values such as 02/08/2026 13:35 are shown without a locale cue, so day/month order is ambiguous.	findings/screenshots/failed/task1_C1_C1-LOCALE-001_01.png	Not submitted
C1-NAV-001	C1	3 — Major accessibility defect	Six sidebar links have no text, aria-label , title , or aria-labelledby value.	findings/screenshots/failed/task1_C1_C1-NAV-001_01.png	Not submitted
C1-SEARCH-001	C1	2 — Minor/medium recovery defect	“No users found matching your filters.” is shown, but there is no visible Clear or next-action control.	findings/screenshots/failed/task1_C1_C1-SEARCH-001_01.png	Not submitted
C2-FORM-001	C2	2 — Minor/medium form clarity defect	First Name displays “Last Name”; Last Name displays “First Name”.	findings/screenshots/failed/task1_C2_C2-FORM-001_01.png	Not submitted
C2-FOCUS-001	C2	3 — Major keyboard/accessibility defect	Focus remains on the background Edit user button while the dialog is open; after Cancel, focus moves to the document body.	findings/screenshots/failed/task1_C2_C2-FOCUS-001_01.png	Not submitted
C2-VALIDATION-001	C2	2 — Minor/medium validation defect	The modal remains open but displays “Last name is required” even though First Name is empty.	findings/screenshots/failed/task1_C2_C2-VALIDATION-001_01.png	Not submitted
C3-CONFIRM-001	C3	3 — Major safety/confirmation defect	The generic Edit User modal presents only Active and Save Changes; no Block/Unblock warning or confirmation is shown before either persistent state change.	findings/screenshots/failed/task1_C3_C3-CONFIRM-001_01.png; findings/screenshots/failed/task1_C3_C3-CONFIRM-001_02.png	Not submitted
C3-FOCUS-001	C3	3 — Major keyboard/accessibility defect	Focus remains on Edit user behind the modal and falls to the document body after Cancel.	findings/screenshots/failed/task1_C3_C3-FOCUS-001_01.png	Not submitted
C3-RESET-001	C3	3 — Major functional gap	The row exposes only Edit user and Delete user; Edit User exposes Close, Cancel, and Save Changes. No Reset Password action is present.	findings/screenshots/failed/task1_C3_C3-RESET-001_01.png	Not submitted

Task 3 subsequently identified the separate physical-phone responsive-layout bug **C1-RESPONSIVE-001**. It is not retroactively counted as a Task 1B failure; the later Android evidence is reconciled in Sections 10 and 11.

The canonical detailed bug and usability records are maintained in the single [bug_usability_findings_log.md](#). Per-cell checklist failure reasons and bug IDs remain in [execution_C1_C2_C3.md](#).

7. Task 2 — User-Testing Plan and Pilot

7.1 Goal and procedure

Starting state: **Admin** → **Users List**.

Find the specified user and verify the intended account. Assign the Event Organizer role, block the account, restore access by unblocking it, and then attempt to reset its password. Tell the moderator when you believe each requested account state has been saved.

The moderator explains that the product—not the participant—is being tested, asks the participant to think aloud, gives no leading hints, and records assistance only when the participant is completely stuck. Informed observation consent and separate recording consent are mandatory.

7.2 Measures

Measure	Operational definition
Completed	All available subtasks achieved/verified without moderator assistance
Partial	Some available subtasks achieved; another available subtask incomplete or assisted
Failed	No meaningful available goal achieved, participant abandons, or moderator performs the available task
Time	Seconds from first measured C1 action to terminal state; system-block time noted separately
Error	Participant action away from goal or requiring recovery; verified system defects excluded
Hesitation	Observable pause beyond a threshold fixed before the pilot, or explicit uncertainty; the threshold was not documented in the supplied plan
Assistance	Each task-relevant moderator intervention

If Reset Password is unavailable, that subtask is **Blocked by system defect** and is not participant failure.

7.3 Neutral probes

- Clarity: Which page or wording was least clear?
- Recovery: What helped or blocked recovery?
- Speed: Which step felt slower or more effortful than expected?
- Trust: When was confidence highest/lowest that EMS saved the right state?
- Improvement: What one change would help most?

7.4 Pilot

The pilot is one separate person and is excluded from P01–P05 aggregates. The student confirmed on 2026-08-04 that the pilot user read and understood the Scenario C flow and achieved a **100% scenario-comprehension pass**. No wording or flow-order change was required.

This confirms comprehension of the test script only; it does not establish live completion of every EMS action. Pilot identity/profile, test date, environment, consent, timing, errors, hesitations, assistance, and recording were not provided. See the [pilot notes](#) and [changes after pilot](#).

8. Task 2 — Five Official Sessions

The student-supplied notes identify five participants and confirm that each is outside 23KTPM1. Contacts are masked and raw recordings were not analysed by AI.

ID	Participant	Outside class	Masked contact	Success	Time	Errors	Hesitations	Assistance	Evidence and concise observation
P01	Nguyen Duy Khanh	Student-confirmed: Yes	033****604	Completed available subtasks	77 s	0	0	0	P01 notes ; Reset Password system-blocked by C3-RESET-001
P02	Mai Hoang Thai	Student-confirmed: Yes	090****533	Completed available subtasks	42 s	0	0	0	P02 notes ; Reset Password system-blocked by C3-RESET-001
P03	Tang Nhat Minh	Student-confirmed: Yes	090****402	Completed available subtasks	50 s	0	0	0	P03 notes ; Reset Password system-blocked by C3-RESET-001
P04	Le Long	Student-confirmed: Yes	037****840	Completed available subtasks	40 s	0	0	0	P04 notes ; Reset Password system-blocked by C3-RESET-001

ID	Participant	Outside class	Masked contact	Success	Time	Errors	Hesitations	Assistance	Evidence and concise observation
P05	Nguyen Ngoc Bao Tram	Student-confirmed: Yes	089****270	Completed available subtasks	91 s	0	0	0	P05 notes ; Reset Password system-blocked by C3-RESET-001

“Completed” applies to every available task step. The full Reset Password goal was completed by 0/5 because the product did not expose the action; this is not participant failure under the predefined rule.

9. Task 2 — SUS, Metrics, Findings, and Recommendations

9.1 SUS scoring

The ten standard statements are preserved in [test_plan.md](#). For each official participant: odd item contribution = response – 1; even item contribution = 5 – response; SUS = contribution sum × 2.5. The pilot is excluded.

Participant	Contribution sum	SUS score
P01	26	65
P02	20	50
P03	30	75
P04	30	75
P05	31	77.5
Mean	27.4	68.5

The earlier draft treated the raw response sum as the contribution sum and therefore did not reverse-score the even items. The values above correct that arithmetic. For example, P02’s response of 4 to every item gives a score of 50, not 80.

The SUS answers provide attitudinal signals and are interpreted alongside the task metrics. Q1, Q3, Q7, and Q9 all average 4.0, supporting positive perceived willingness, ease, learnability, and confidence. The clearest caution is clumsiness (Q8 mean 3.4), followed by a mixed inconsistency signal (Q6 mean 2.8). P02 answered 4 to every positive and negative statement; its score of 50 is therefore interpreted cautiously rather than mapped to a specific UI cause. Full participant/item derivation is in [sus_responses.md](#).

9.2 Aggregate task metrics

Metric	Result
Available-task success rate	5/5 = 100%
Full Reset Password completion	0/5 = 0% (system-blocked)
Mean time	$(77 + 42 + 50 + 40 + 91) \div 5 = 60.0$ seconds
Mean errors	0.0
Mean hesitations	0.0
Mean assistance	0.0
Mean SUS	68.5 / 100

Raw de-identified response and formula tables are in [sus_responses.md](#) and [metrics_summary.md](#).

9.3 Task 2 conclusion

All five participants were unable to complete Reset Password because no such action was exposed. Evidence consists of the [five session notes](#) and the [genuine SUT screenshot](#). This is the Task 2 impact of the existing functional bug [C3-RESET-001](#), not an additional bug or separate usability-finding ID. The supplied notes contain no structured friction observations or probe answers supporting another conclusion.

9.4 Prioritised recommendations

- P0 — Restore the missing control.** Add a permission-controlled Reset Password action in a predictable location on the selected user’s row or edit/detail flow.
- P0 — Make the action safe and trustworthy.** Use an account-specific confirmation with a safe default, explain the consequence, avoid disclosing credentials, show clear success/failure feedback, and record an audit event.

3. **P1 — Validate the repair.** Retest C3 with real users and capture structured think-aloud notes plus the clarity, recovery, speed, trust, and improvement probe answers.

The detailed Task 2 conclusion is in [usability_findings.md](#). **C3-RESET-001** remains the single canonical row in the [Consolidated Bug & Usability Findings Log](#).

9.5 Remaining submission action

- **C3-RESET-001** still requires manual Google Form submission and a real timestamp/receipt.

10. Task 3 — Cross-Browser / Cross-Platform

Every screen uses the same five planned configurations:

1. Windows + Edge + Desktop
2. Windows + Firefox + Desktop
3. Windows + Opera + Desktop
4. iPadOS + Safari + Tablet
5. Android + Chrome + Phone

Cell	Screen	OS	Browser	Class	Device; date; environment	Result	Screenshot
C1-01	C1	Windows	Edge	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C1-01
C1-02	C1	Windows	Firefox	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C1-02
C1-03	C1	Windows	Opera	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C1-03
C1-04	C1	iPadOS	Safari	Tablet	iPad Air M6; 2026-08-04; Physical	PASS	C1-04
C1-05	C1	Android	Chrome	Phone	Samsung Galaxy A70; 2026-08-04; Physical	FAIL	C1-05
C2-01	C2	Windows	Edge	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C2-01
C2-02	C2	Windows	Firefox	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C2-02
C2-03	C2	Windows	Opera	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C2-03
C2-04	C2	iPadOS	Safari	Tablet	iPad Air M6; 2026-08-04; Physical	PASS	C2-04
C2-05	C2	Android	Chrome	Phone	Samsung Galaxy A70; 2026-08-04; Physical	PASS	C2-05
C3-01	C3	Windows	Edge	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C3-01
C3-02	C3	Windows	Firefox	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C3-02
C3-03	C3	Windows	Opera	Desktop	MSI GF63; 2026-08-04; Physical	PASS	C3-03
C3-04	C3	iPadOS	Safari	Tablet	iPad Air M6; 2026-08-04; Physical	PASS	C3-04
C3-05	C3	Android	Chrome	Phone	Samsung Galaxy A70; 2026-08-04; Physical	PASS	C3-05

Overall result: **14 PASS and 1 FAIL**. **C1-RESPONSIVE-001** records horizontal overflow and severe compression on the Android/Chrome phone cell. The C3 PASS values apply only to the available Edit User/Active compatibility path; Reset Password remains unavailable as the already confirmed functional gap **C3-RESET-001**.

Coverage proof

Screen	OS (3)	Browsers (5)	Classes (3)	Planned	Executed
C1	Windows, iPadOS, Android	Edge, Firefox, Opera, Safari, Chrome	Desktop, Tablet, Phone	5	5
C2	Windows, iPadOS, Android	Edge, Firefox, Opera, Safari, Chrome	Desktop, Tablet, Phone	5	5
C3	Windows, iPadOS, Android	Edge, Firefox, Opera, Safari, Chrome	Desktop, Tablet, Phone	5	5

All 15 cells have real screenshots showing the student email overlay, EMS URL, browser, OS, and device identity. See the complete [compatibility matrix](#).

11. Consolidated Bug & Usability Findings Log

The unified log contains nine confirmed Task 1B bugs and one confirmed Task 3 compatibility/UI bug. Task 2 adds no duplicate row: its 5/5 system-block conclusion corroborates **C3-RESET-001**. Google Form timestamps and confirmation references remain unclaimed until the student submits the Form manually.

ID	Screen	Type	Description	Evidence	Form status	Overall status
C1-LOCALE-001	Scenario C / C1 Users List	Localisation / usability defect	Created and Updated dates use an ambiguous numeric format in the English UI.	findings/screenshots/failed/task1_C1_C1-LOCALE-001_01.png	Not submitted	Confirmed — pending Google Form submission
C1-NAV-001	Scenario C / C1 Users List	Accessibility / navigation defect	Six sidebar destination links have no accessible names.	findings/screenshots/failed/task1_C1_C1-NAV-001_01.png	Not submitted	Confirmed — pending Google Form submission
C1-SEARCH-001	Scenario C / C1 Users List	Usability / recovery defect	The no-result search state offers no Clear or next action.	findings/screenshots/failed/task1_C1_C1-SEARCH-001_01.png	Not submitted	Confirmed — pending Google Form submission
C1-RESPONSIVE-001	Scenario C / C1 Users List	Responsive/mobile UI defect	Users List overflows horizontally and is severely compressed on Android/Chrome phone.	task3_compatibility/screenshots/C1/C1-05_C1_Android_Chrome_Phone_20260804.jpg	Not submitted	Confirmed — pending Google Form submission
C2-FORM-001	Scenario C / C2 Assign Role/Edit User	Form clarity defect	First Name and Last Name placeholders are reversed.	findings/screenshots/failed/task1_C2_C2-FORM-001_01.png	Not submitted	Confirmed — pending Google Form submission
C2-FOCUS-001	Scenario C / C2 Assign Role/Edit User	Accessibility / modal focus defect	Edit User does not move focus into the modal or restore it after Cancel.	findings/screenshots/failed/task1_C2_C2-FOCUS-001_01.png	Not submitted	Confirmed — pending Google Form submission
C2-VALIDATION-001	Scenario C / C2 Assign Role/Edit User	Validation defect	Empty First Name produces a Last Name required error.	findings/screenshots/failed/task1_C2_C2-VALIDATION-001_01.png	Not submitted	Confirmed — pending Google Form submission
C3-CONFIRM-001	Scenario C / C3 Block/Unblock	Safety / confirmation defect	Active-state changes have no Block/Unblock warning or confirmation.	findings/screenshots/failed/task1_C3_C3-CONFIRM-001_01.png; findings/screenshots/failed/task1_C3_C3-CONFIRM-001_02.png	Not submitted	Confirmed — pending Google Form submission
C3-FOCUS-001	Scenario C / C3 Block/Unblock	Accessibility / modal focus defect	The Block/Unblock modal path does not move or restore focus.	findings/screenshots/failed/task1_C3_C3-FOCUS-001_01.png	Not submitted	Confirmed — pending Google Form submission

ID	Screen	Type	Description	Evidence	Form status	Overall status
C3-RESET-001	Scenario C / C3 Reset Password	Functional gap	Reset Password is absent from the dedicated user's available actions.	findings/screenshots/failed/task1_C3_C3-RESET-001_01.png	Not submitted	Confirmed — pending Google Form submission

The detailed log also contains preconditions, reproduction steps, expected and actual results, suggested fixes, source-task references, reporter email, and submission status.

12. Agent Skills and Demonstration

12.1 EMS checklist executor

[SKILL.md](#) accepts the SUT/authorization, scenario/screens, checklist/output/evidence paths, and tester environment. It prepares, executes, or audits every item while requiring live evidence for Passed/Failed. Its validator checks IA coverage, row completeness, statuses, failure metadata, and duplicate IDs; it never decides UI quality.

12.2 Record AI audit

[SKILL.md](#) accepts real interaction metadata, the verbatim prompt, and the verbatim AI output or a labelled artefact reference. It blocks likely secrets/PII, allocates stable IDs, rejects duplicates, writes the four required fields, and updates the prompt log and audit-report index.

12.3 Demonstration scope

Both skills are repository-local submission artefacts and are not globally installed. They cannot replace human EMS observation, participant recruitment, or Google Form actions. Only the [ems-checklist-executor](#) demonstration is required. Its video link is tracked in [demo_video_links.md](#) and will be added after the student records the demo.

12.4 Validation results

Check	Result
Current ems-checklist-executor final validation	OK: 56 checklist rows, 56 execution rows, 168 cells, 150 Passed, 18 Failed, 0 working blockers
record-ai-audit privacy/schema dry-run for AI-010	Passed
record-ai-audit entry/index generation for AI-010	Passed
Task 2 numerical recomputation	Passed: 300 total seconds, 60.0-second mean, 68.5 mean SUS
Checked local Markdown links in changed report artefacts	All resolved
git diff --check	Passed; line-ending conversion warnings only

13. AI Use and Human Review

Codex was used for requirement comparison, scaffold generation, workbook content recovery and Markdown migration, skill/script creation, the authorised Task 1B live browser run and evidence capture, deterministic consistency checks, and recalculation/cross-reconciliation of the student-supplied de-identified Task 2 metrics and SUS responses.

Codex did not recruit participants, answer SUS on their behalf, open or analyse the participant recordings, invent probe answers or pilot evidence, perform the student's physical-device Task 3 runs, or submit Google Forms.

Mandatory assignment declaration:

I use AI tools for the following tasks,

The disclosure and factual interaction index are in [ai_audit_report.md](#), and verbatim interaction records are in [prompt_log.md](#).

13.1 AI Critique

The 200–300 word student critique documents the incorrect initial SUS calculation, duplicate [UF-001](#), Task 3 mobile-overflow consolidation, excessive placeholders, and the audit-schema fields removed during human review. See [ai_critique.md](#).

14. Conclusion and Self-Assessment

All planned testing content is complete and synchronized: the shared checklist contains 56 approved items; Task 1B executed all 168 screen-item cells; Task 2 documents one pilot and five official participants; and Task 3 executed all 15 compatibility cells. The consolidated result is 10 verified bugs, including the mobile overflow defect **C1-RESPONSIVE-001**, and no duplicate Task 2 usability finding for the already recorded Reset Password defect **C3-RESET-001**.

No.	Criterion	Maximum	Self-assessment
1a	Task 1A — Shared checklist, sources, and AI prompts	15	15
1b	Task 1B — Checklist execution and bug reports	15	15
2	Task 2 — User testing and usability report	25	25
3	Task 3 — Cross-browser/cross-platform testing	25	25
4	Google Form findings and aggregated log	10	10
5	Agent Skills	10	10
Total		100	100

The claimed score is finalized after the five submission actions listed at the end of this report.

15. Appendices

Appendix A — AI Audit Report Snapshot

Four interactions are indexed: AI-007 (scaffold), AI-008 (checklist approval update), AI-009 (spreadsheet removal after Markdown migration), and AI-010 (Task 2 analysis and audit). Each record contains only the AI tool name, date/time, prompt, and AI output or a clearly labelled output reference. The authoritative detailed appendix is [AI/ai_audit_report.md](#).

Appendix B — Prompt Log Snapshot

- AI-007: current Vietnamese request to complete HW03 from supplied materials and the master prompt, recorded 2026-08-01T19:05:18+07:00; artefact output references recorded.
- AI-008: student confirmation that the group reviewed and approved all 56 checklist items, recorded 2026-08-01T19:58:58+07:00; synchronized status references recorded.
- AI-009: student request to keep the working package in Markdown and remove Excel files after migration verification, recorded 2026-08-01T20:20:25+07:00.
- AI-010: student request to finish Task 2 from five supplied session records and audit the result with a skill, recorded 2026-08-03T22:41:45+07:00; affected artefact references recorded without participant PII or recording content.

See [AI/prompt_log.md](#).

Appendix C — Git Commit Log

The student has intentionally deferred generation of the submission commit log until all HW03 content is complete. See [git_commit_log.txt](#).

Appendix D — Evidence Index and Demo Links

Evidence group	Path	Status
Shared checklist/review	group/	Approved 56/56 unchanged
Task 1 failed screenshots	findings/screenshots/failed/	10 genuine SUT PNG files present; 8 unique hashes because individual captures support more than one finding
Task 2 official sessions	task2_user_testing/sessions/	Five participant notes, metrics, and recording references supplied
Task 2 metrics/SUS/conclusion	task2_user_testing/metrics_summary.md, sus_responses.md, usability_findings.md	Aggregates corrected; 5/5 system-block conclusion linked to existing bug C3-RESET-001 without a duplicate finding ID
Task 2 pilot	task2_user_testing/pilot/	100% scenario-comprehension pass documented; no flow wording change required

Evidence group	Path	Status
Compatibility screenshots	task3_compatibility/screenshots/C1/, C2/, C3/	15 executed cells and 15 distinct JPG files; 14 Passed, 1 Failed
Google Form	External submission	Pending completion
Skill demo	agent/demo_video_links.md	One ems-checklist-executor video link pending
Git commit log	git_commit_log.txt	Generate after content finalization
Final PDFs	main_report.pdf , AI/ai_audit_report.pdf	Export after final review

The only remaining actions are: complete the Google Form, record and add the [ems-checklist-executor](#) demo link, generate the final Git commit log, export the required PDFs, then create the ZIP and submit the package.